

A Complete- F Self-Gated Power-Mean–Anchored Hill- q Rule for Long-Peptide Aggregation

IRIS methodological development memo

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Abstract

IRIS assigns Level 1 scores to candidate minimal-epitope–HLA tuples, whereas the PDAC and SARS-CoV-2 assays label unresolved long peptides. The first self-gated Hill- q proposal was anchored at the strongest candidate and could therefore express only scores between maximum and log-sum-exp. New full-roster tournament output suggested that COVID favors below-maximum averaging, but those outputs also exposed a more fundamental problem: candidates omitted from Level 1 computation had been represented by the numerical floor. The apparent preference therefore partly measures enumeration coverage rather than biology. This revision makes complete Level 1 tensors a prerequisite, replaces the maximum anchor by a generalized power-mean anchor, retains floor-aware Hill corroboration and the self-consistent gate, and makes worst-cohort rank the primary cross-context calibration objective. No production parameter grid is frozen here. The complete- F rerun and clean-data diagnosis must occur first.

1 Why Level 2 aggregation is needed

1.1 The experimental unit is a long peptide

For a candidate minimal epitope e_i and restricting HLA allele h_i , IRIS produces a nonnegative mechanistic score

$$F_i = F(e_i, h_i). \quad (1)$$

The definition and calibration of F belong to Level 1. The Level 2 problem arises because a measured long peptide ℓ generally contains a roster

$$\mathcal{C}_\ell = \{(e_i, h_i) : i = 1, \dots, K_\ell\} \quad (2)$$

of candidate minimal-epitope–HLA tuples, while the observed PDAC or COVID response label belongs to ℓ . The experiment does not identify which member of $\mathcal{C}_\ell$, if any, caused the response. Level 2 must therefore map the complete candidate-score roster to one endpoint score:

$$S_\ell = S(F_1, \dots, F_{K_\ell}). \quad (3)$$

Because the Level 1 score is nonnegative and multiplicative, the computation uses the log scale

$$z_i = \log(F_i + \epsilon), \quad \epsilon = 10^{-12}. \quad (4)$$

This transformation stabilizes products and makes score ratios additive. Neither F_i nor $\text{sigmoid}(z_i)$ is a response probability. The aggregate is a ranking score on the Level 1 scale, not a calibrated probability of immunogenicity.

1.2 Three aggregation questions

Three plausible biological configurations motivate the design.

One-candidate sufficiency. One candidate may be strong enough to explain the long-peptide response. A rule should be able to protect such a leader from dilution by weak roster members.

Leader skepticism. An isolated high score may be less persuasive when the rest of the roster is weak. A rule should also be able to average below the leader when the component/cohort geometry supports that behavior.

Multi-candidate corroboration. Several consequential candidates may jointly provide persuasive evidence even when none is individually decisive. Their magnitudes should be allowed to accumulate, but not merely because more computational opportunities were enumerated.

Three familiar fixed rules isolate these questions:

$$S_{\max}(z) = \max_i z_i, \quad (5)$$

$$S_{\text{mean}}(z) = \frac{1}{K_\ell} \sum_i z_i, \quad (6)$$

$$S_{\text{LSE}}(z) = \log \sum_i e^{z_i}. \quad (7)$$

Maximum protects a leader but discards all corroboration. Mean discounts an isolated leader but can be dominated by genuinely weak candidates. Log-sum-exp accumulates evidence but rises with candidate opportunity. The revised adaptive family first chooses an anchor on the mean-to-maximum axis, then offers corroboration toward a complete accumulation ceiling, and finally lets the developing aggregate gate that offer.

1.3 Roster geometry and why coverage matters

Candidate opportunity differs before any score is computed. A peptide of length $L \geq 12$ contains

$$W(L) = \sum_{r=9}^{12} (L - r + 1) = 4L - 38 \quad (8)$$

sequence windows of lengths 9–12 before HLA expansion. Thus a 15-mer offers 22 sequence windows while a 27-mer offers 70. Pairing those windows with a patient’s distinct HLA alleles expands the tuple roster further. Peptide length, HLA coverage, homozygosity, overlapping frames, and preprocessing decisions can therefore change the opportunity set seen by Level 2.

This does not mean that roster size should automatically increase or decrease the score. It means that the roster convention must be explicit and that every declared tuple must be treated consistently. Exact duplicate biological tuples are removed; distinct minimal epitopes or distinct HLA restrictions remain distinct. The formula does not claim that the remaining contributions are statistically independent.

1.4 What the recent outputs do and do not establish

The newer full-roster tournament produced two useful warnings. First, under the candidate-conservative view, the max-anchored adaptive systems had no survivors, and COVID appeared to favor below-maximum averaging. Second, a large share of the purported full roster had never received a Level 1 tensor calculation and was assigned $\log \epsilon$ by omission.

The second fact prevents the first observation from being interpreted directly. Mean and related operators were partly measuring how many windows the pipeline skipped. The apparent COVID preference, the zero-survivor result, and any component-specific magnitudes from that run are therefore provisional. They motivate a complete- F rerun, not a production alpha grid.

One structural conclusion survives this contamination: a family constrained to $[\max, \text{LSE}]$ cannot express below-maximum behavior on *any* dataset. Adding a power-mean anchor repairs that expressive limitation without fitting its parameter values to the contaminated outputs. Whether complete- F COVID still prefers a low alpha is an empirical question deliberately left open.

2 Decision and sequencing

The current omission-floor full-roster results are not valid calibration data for a new Level 2 rule. A floor can be interpreted biologically only when the corresponding Level 1 calculation was actually performed. It cannot stand for an uncomputed value. Work therefore proceeds in this order:

1. Freeze the existing Level 1 geometry, decision regime, exposure grid, and NCI-selected settings. Do not reselect Level 1.
2. Compute the Level 1 F tensor for every declared COVID and PDAC 9–12-mer–HLA tuple.
3. Deduplicate the roster as a set of (endpoint, minimal epitope, normalized HLA) tuples.
4. Rebuild and audit the transfer package. Every retained mapping row must resolve to one computed tensor score.
5. Re-diagnose component and Level 2 preferences on complete- F data.
6. Only then freeze the α grid and jointly select (α, q, c, κ) .
7. Enter the frozen selected systems into the later candidate-conservative tournament.

The omission-floor run may be retained as a sensitivity analysis documenting coverage-by-omission. It must not supply production parameter choices or be described as if its inputs were complete.

3 What changed in the model

3.1 Why the maximum anchor was insufficient

Let z_i be the log-scale Level 1 score for candidate i . The former rule started at $m = \max_i z_i$ and admitted a nonnegative corroboration bonus. It was therefore bounded by

$$m \leq S \leq \text{LSE}(z).$$

No choice of Hill order or gate parameters could reproduce mean behavior below m . That was a structural limitation independent of the data-contamination problem.

The power-mean anchor introduces the missing downward degree of freedom while retaining the self-gated corroboration mechanism. On the log-score scale, $\alpha = 0$, $\alpha = 1$, and $\alpha = \infty$ recover the arithmetic mean, log-mean-exp, and maximum, respectively.

3.2 Complete- F input contract

For a long peptide ℓ , let

$$\mathcal{C}_\ell = \{(e_i, h_i) : i = 1, \dots, K_\ell\}$$

be the set of unique minimal-epitope–HLA tuples after the 9–12-mer length restriction. Overlapping frames and different HLA restrictions remain distinct tuples; duplicate observations of the same tuple enter once.

Let $F_i \geq 0$ be the computed Level 1 signal and fix

$$\epsilon = 10^{-12}, \quad z_i = \log(F_i + \epsilon).$$

A value $z_i = \log \epsilon$ means that a completed Level 1 calculation produced zero signal. A missing calculation has no z_i and makes the input package invalid. This distinction is enforced by the implementation.

4 Power-mean anchor

The anchor answers the first Level 2 question: how much should the strongest candidate dominate before corroboration is considered? It is defined on the positive scale and then reported on the log scale. For $\alpha > 0$,

$$e^{A_\alpha} = \left(\frac{1}{K_\ell} \sum_i (e^{z_i})^\alpha \right)^{1/\alpha}.$$

Taking the logarithm gives the numerically convenient form below.

For finite $\alpha > 0$, define

$$A_\alpha(z) = \frac{1}{\alpha} \log \left(\frac{1}{K_\ell} \sum_{i=1}^{K_\ell} e^{\alpha z_i} \right). \quad (9)$$

The continuous endpoint and maximum limit are

$$A_0(z) = \frac{1}{K_\ell} \sum_i z_i, \quad (10)$$

$$A_\infty(z) = \max_i z_i. \quad (11)$$

The family therefore includes

$$A_0 = \text{mean}(z), \quad (12)$$

$$A_1 = \log \left(\frac{1}{K_\ell} \sum_i e^{z_i} \right) = \text{logmeanexp}(z), \quad (13)$$

$$A_\infty = \max(z). \quad (14)$$

The order is monotone:

$$A_0 \leq A_\alpha \leq A_\beta \leq A_\infty \quad \text{whenever} \quad 0 \leq \alpha \leq \beta.$$

Increasing alpha therefore gives progressively more authority to the upper tail. It does not change the candidate roster and it does not inspect an endpoint label.

Anchor order	Interpretation on the log-score roster
$\alpha = 0$	Arithmetic mean of z ; an isolated leader can be strongly discounted by the remaining roster.
$0 < \alpha < 1$	Smooth interpolation between mean and log-mean-exp.
$\alpha = 1$	Log-mean-exp; positive-scale arithmetic mean of e^z .
$1 < \alpha < \infty$	Increasingly upper-tail-sensitive soft maximum.
$\alpha = \infty$	Exact maximum; reproduces the former anchor.

This anchor is not yet the final aggregate. A low alpha permits descent below the strongest candidate, while the corroboration stage can still move the score upward when several completed Level 1 signals support that move. The roles of alpha and q are therefore different: alpha controls the baseline attitude toward the leader; q controls how the distribution of positive signal counts as corroborative breadth.

Only nonnegative α is proposed for the first complete- F evaluation. Negative orders would give increasing leverage to near-zero candidates and should not enter without separate scientific justification. The production set of nonnegative orders is intentionally not specified here.

5 Floor-aware Hill corroboration

5.1 Signal weights

Recover the floor-free Level 1 signal by

$$x_i = \max(e^{z_i} - \epsilon, 0). \quad (15)$$

When $X = \sum_i x_i > 0$, set $p_i = x_i/X$. Candidates whose completed Level 1 signal is exactly zero carry no Hill mass. This prevents a collection of true numerical floors from manufacturing breadth.

The p_i sum to one and describe how the completed Level 1 signal is distributed across candidates. They are compositional weights, not response probabilities and not probabilities that a particular epitope is causal.

5.2 Hill effective number

For order $q \geq 0$,

$$N_q(p) = \begin{cases} \left(\sum_{i:p_i>0} p_i^q \right)^{1/(1-q)}, & q \neq 1, \\ \exp \left(- \sum_{i:p_i>0} p_i \log p_i \right), & q = 1, \end{cases} \quad (16)$$

with

$$N_0 = \#\{i : x_i > 0\}, \quad N_\infty = \frac{1}{\max_i p_i}.$$

Define

$$D_q = 1 - \frac{1}{N_q}. \quad (17)$$

This breadth fraction is zero for a single positive-signal candidate. For K_+ equal positive signals it is $1 - 1/K_+$ at every order. The orders differ only when the roster is uneven. Hill numbers are nonincreasing in q : low orders recognize more of the positive roster as breadth, while high orders require mass to remain close to the leader.

Hill order	Breadth interpretation
$q = 0$	Richness-sensitive: every strictly positive completed signal counts, regardless of its magnitude.
$q = 1$	Entropy-weighted: candidates contribute according to information mass.
$q = 2$	Concentration-sensitive: $D_2 = 1 - \sum_i p_i^2$.
$q = \infty$	Leader-share-sensitive: $D_\infty = 1 - \max_i p_i$.

At $q = 0$, adding a tiny but positive signal changes richness. That is an explicit modeling choice, not an invariance claim. Higher orders reduce this sensitivity, while exact zeros contribute at no order.

5.3 Available accumulation

The complete accumulated ceiling is

$$U = \log \left(\epsilon + \sum_i x_i \right). \quad (18)$$

For every nonempty roster and $\alpha \geq 0$,

$$A_\alpha \leq \max_i z_i \leq U.$$

The offer available above the chosen anchor is

$$C_{\alpha,q} = D_q \max(U - A_\alpha, 0). \quad (19)$$

Thus α chooses how skeptical the baseline is about an isolated leader, while q chooses how concentrated the corroborating signal may be.

The construction can be read in three stages:

1. $U - A_\alpha$ asks how much complete accumulation is available above the chosen anchor.
2. D_q asks what fraction of that availability is justified by the distribution of positive signal.
3. $C_{\alpha,q}$ is only an *offer*; the gate below decides how much enters the final score.

For a singleton positive-signal roster, $N_q = 1$, $D_q = 0$, and the offer is zero. For an all-zero completed roster, the computation exits before normalization and returns the declared floor. These are different cases: the first has one informative candidate, while the second has a completed calculation but no positive Level 1 signal.

6 Self-consistent gate

For center $c \in \mathbb{R}$ and scale $\kappa > 0$, define the aggregate as the solution of

$$S = \underbrace{A_\alpha}_{\text{power-mean anchor}} + \underbrace{C_{\alpha,q}}_{\text{corroboration offer}} \times \underbrace{\text{sigmoid}\left(\frac{c-S}{\kappa}\right)}_{\text{fraction admitted}}. \quad (20)$$

The gate admits more of the offer when the resulting score remains below the center and less when the result is already strong.

For a provisional aggregate a , write

$$g_{c,\kappa}(a) = \text{sigmoid}\left(\frac{c-a}{\kappa}\right) = \frac{1}{1 + \exp[(a-c)/\kappa]}. \quad (21)$$

The center c is the aggregate score at which one-half of the offer is admitted. The positive width κ controls how gradual the transition is: the gate falls from 0.9 to 0.1 over $2 \log(9)\kappa \approx 4.39\kappa$ score units.

The definition is implicit by design. Evaluating the gate only at the anchor would ignore the score increase caused by admitted corroboration. Here a large offer raises S , and the increase in S closes the same gate through which the offer enters. This negative feedback is the self-gating mechanism.

Quantity	Question answered
A_α	How much authority should the roster give its strongest candidate before corroboration?
$U - A_\alpha$	How much complete positive-scale accumulation is available above that anchor?
D_q	How should the concentration of positive signal count as effective breadth?
$C_{\alpha,q}$	How much corroboration is offered before sufficiency gating?
$g_{c,\kappa}(S)$	What fraction of the offer does the resulting aggregate still need?

Let

$$f(S) = S - A_\alpha - C_{\alpha,q} \text{sigmoid}\left(\frac{c-S}{\kappa}\right).$$

At the interval endpoints,

$$f(A_\alpha) \leq 0, \quad f(A_\alpha + C_{\alpha,q}) \geq 0.$$

Moreover,

$$f'(S) = 1 + \frac{C_{\alpha,q}}{\kappa} \text{sigmoid}\left(\frac{c-S}{\kappa}\right) \left[1 - \text{sigmoid}\left(\frac{c-S}{\kappa}\right)\right] > 0.$$

Therefore Equation (20) has exactly one solution and certified bisection may be used on

$$A_\alpha \leq S \leq A_\alpha + C_{\alpha,q} \leq U. \quad (22)$$

If $K_\ell = 0$, or if every completed signal is zero, define

$$S = \log \epsilon, \quad C_{\alpha,q} = 0.$$

An empty biological roster is allowed; an incomplete tensor roster is not.

7 How the rule behaves

7.1 Certified bounds and monotonicity

Equation (22) says more than that the solver is numerically safe. It locates the scientific behavior of the score:

$$A_\alpha \leq S \leq A_\alpha + C_{\alpha,q} \leq U.$$

The aggregate never falls below its declared power-mean anchor and never exceeds complete positive-scale accumulation. At $\alpha = \infty$, this becomes the former max-to-accumulation interval. At $\alpha = 0$, the interval begins at the arithmetic mean of the log scores and therefore includes the missing below-maximum regime.

For fixed anchor and gate parameters, increasing the offer cannot lower the solution. Implicit differentiation gives

$$\frac{\partial S}{\partial C} = \frac{g_{c,\kappa}(S)}{1 - Cg'_{c,\kappa}(S)} > 0. \quad (23)$$

This property is one reason to gate at the solved score rather than evaluate an explicit shortcut at the fully accumulated endpoint. Alpha changes both the anchor and the available offer, however, so the final score need not be monotone in alpha for every roster and gate setting.

7.2 Limiting cases

Limit or roster	Consequence
One positive candidate	$D_q = 0$, so $S = A_\alpha = z_1$ for every alpha, q , c , and κ .
All completed signals zero	$S = \log \epsilon$ and no Hill distribution is formed.
$\alpha = 0$	The anchor is $\text{mean}(z)$; the family can discount an isolated leader.
$\alpha = 1$	The anchor is $\text{logmeanexp}(z)$.
$\alpha = \infty$	The anchor is maximum; this recovers the expressive range of the former proposal, with corrected floor semantics.
$c \rightarrow -\infty$	The gate closes and $S \rightarrow A_\alpha$.
$c \rightarrow +\infty$	The gate opens and $S \rightarrow A_\alpha + C_{\alpha,q}$.
$\kappa \rightarrow \infty$	The gate tends to one-half and $S \rightarrow A_\alpha + C_{\alpha,q}/2$.

As κ approaches zero, the transition becomes nearly hard:

$$S \approx \begin{cases} A_\alpha, & A_\alpha \geq c, \\ A_\alpha + C_{\alpha,q}, & A_\alpha + C_{\alpha,q} \leq c, \\ c, & A_\alpha < c < A_\alpha + C_{\alpha,q}. \end{cases} \quad (24)$$

A strong anchor therefore receives essentially no bonus; a weak roster whose full offer remains below c can receive nearly all of it; and an offer that would cross c is admitted only until the developing aggregate reaches the transition region. Finite κ joins these cases smoothly.

7.3 Worked score geometries

The following examples use log scores far above the numerical floor so that $x_i \approx e^{z_i}$. They illustrate the decomposition before any production parameters are chosen.

Roster	Key quantities	Interpretation
Singleton (-2)	$A_\alpha = -2; D_q = 0$	The score is exactly -2 . A singleton is coherent under every setting.
Three-way weak plateau $(-2, -2, -2)$	$A_\alpha = -2;$ $U \approx -0.901;$ $D_q = 2/3$	Alpha and q are irrelevant on an equal plateau. The offer is about 0.732, so the gate selects a score in $[-2, -1.268]$.
Isolated leader $(0, -5, -5)$	$A_0 = -3.333;$ $A_1 \approx -1.085;$ $A_\infty = 0;$ $U \approx 0.013$	Alpha controls whether the leader is discounted. At $q = 2$ the two weak signals provide little breadth; at $q = 0$ their positive richness counts much more strongly.
Three-way strong plateau $(0, 0, 0)$	$A_\alpha = 0;$ $U \approx 1.099;$ $D_q = 2/3$	The offer is substantial, but a gate centered below zero can admit only a small fraction because the anchor is already strong.
All-zero completed roster	$z_i = \log \epsilon; x_i = 0$	The score remains exactly $\log \epsilon$. Repeated numerical floors cannot manufacture corroboration.

The isolated-leader example displays why both order parameters are needed. Alpha determines the starting point, while q determines whether the weak positive tail constitutes breadth. Neither parameter alone spans both questions. The examples also show why complete- F matters: replacing uncomputed entries by $\log \epsilon$ would change the low-alpha anchors and could turn a coverage pattern into an apparent biological geometry.

8 Calibration and evidence

8.1 Joint parameter

The unit of calibration is the complete tuple

$$(\alpha, q, c, \kappa).$$

Parameters may be selected separately for each component model and metric branch, but each selected tuple is then fixed across every cohort in the policy. Component choice is not hidden inside the Level 2 fit.

8.2 Freeze one joint grid

Selection evaluates one declared Cartesian grid

$$\mathcal{G} = \mathcal{A} \times \mathcal{Q} \times \mathcal{C} \times \mathcal{K} \tag{25}$$

for each component-metric group. Alpha is not screened in a preliminary stage and then held fixed while the remaining parameters are optimized; doing so would understate the selection search. Every tuple competes in the same cohort-specific ranking surface.

The complete grid must be frozen before its empirical surfaces are examined. Adding intermediate alpha, q , c , or κ values after viewing a rank-based objective changes that objective through grid

density. The selector therefore has no default alpha list: a production invocation must name the preregistered values explicitly.

Outputs should retain the complete surfaces, the selected tuple, the alpha and Hill orders represented among near-optimal tuples, and boundary flags for c and κ . A selected endpoint order such as zero or infinity is a scientifically interpretable member of the frozen set, not an instruction to extend the grid after the fact.

8.3 Worst-cohort-first objective

For each cohort, rank every eligible tuple by the policy metric and convert to fractional rank. For the all-context policy, select by:

1. minimum worst-cohort fractional rank;
2. minimum equal-cohort mean fractional rank;
3. minimum equal-cohort mean metric regret;
4. declared evidence and numerical tie-breaks.

This maximin ordering matches the fact that the candidate-conservative criterion is defeated by a weak cohort, not rescued by an excellent average. For the PDAC-only policy the first two quantities reduce to PDAC rank; COVID remains a frozen transport assessment.

For PR, a tuple remains eligible only when the staged paired-CNAP comparison supports it directionally over the same component model’s fixed maximum in every policy cohort. ROC uses the same worst-cohort-first ordering on AUROC. These are parameter-selection rules, not a substitute for the later complete candidate-conservative tournament.

8.4 Transport gate

Leave-one-cohort-out results must be generated before production selection. The earlier memo treated them only descriptively. A quantitative transport failure threshold should be preregistered after the complete- F diagnostic and before the production grid is evaluated. This memo does not invent that threshold from contaminated outputs.

8.5 Label blindness and evidential status

Once a tuple is selected, endpoint adaptation is label blind. The endpoint’s candidate scores determine A_α , D_q , $C_{\alpha,q}$, and the solved score. Neither the response label nor the cohort identity chooses a different alpha, Hill order, or gate for that endpoint.

Labels do participate in model-development selection of (α, q, c, κ) . If the same PDAC and COVID cohorts later supply the directed round robin, that tournament is conditional post-selection evidence, not independent external confirmation. The selection grid, complete surfaces, eligibility outcomes, solver contract, input checksums, and this evidential status belong in the manifest.

8.6 Component-level scope

Level 2 selection occurs within each component construction. It can change how that component’s candidate scores are pooled, but it cannot transform a Full-HLA score geometry into a Mono-Q/full-PN geometry or vice versa. A component-level context reversal may therefore remain after the best Level 2 rule is chosen. The production report should distinguish failure of the aggregator to transfer from failure of the underlying component family to transfer.

9 Design requirements

The equations are compact, but they answer only part of the design problem. The following requirements make the roster, numerical, and evidential assumptions explicit.

P1. Complete computation.

Every declared roster tuple has a Level 1 tensor result. Missing values fail closed.

P2. Set semantics.

The roster is a set of endpoint–minimal-epitope–HLA tuples. Exact duplicates do not create breadth.

P3. Downward and upward reach.

The family contains mean, log-mean-exp, and maximum anchors and may admit corroboration up to the complete accumulation ceiling.

P4. One-candidate safety.

A singleton positive-signal roster has $D_q = 0$, so $S = A_\alpha = z_1$.

P5. Zero-floor safety.

Completed zero signals carry no Hill mass. An all-zero roster cannot manufacture corroboration.

P6. No implicit multiplicity claim.

Distinct overlapping windows and HLA tuples may contribute, but the score does not assert their biological independence.

P7. Certified numerics.

The unique root is solved only within Equation (22); nonconvergence fails closed.

P8. Robust calibration.

Cross-context selection is worst-cohort-first, with the full tuple selected jointly and fixed after selection.

P9. Honest scope.

Level 2 adaptivity cannot repair a component family that is structurally wrong for a cohort. Component-level reversals are reported explicitly.

9.1 Candidate-opportunity comparability

Roster richness is neither automatically supportive nor automatically suspect. It may arise from peptide length, HLA coverage, genuine biological breadth, or computational enumeration. The revised rule exposes rather than hides its sensitivity:

- low alpha anchors respond to the complete roster, including weak completed candidates;
- $q = 0$ treats every positive signal as richness;
- higher q values require more signal mass near the leader;
- the accumulation gap and self-gate limit how far corroboration can raise the anchor;
- exact zero signals contribute no Hill mass.

Because the low-alpha anchor is intentionally roster-sensitive, complete enumeration is not merely a data-cleaning preference; it is an identification condition for the model. Two endpoints cannot be compared fairly if one received complete Level 1 calculations and the other had uncomputed candidates inserted at the floor.

The rule does not add a separate penalty based only on K_ℓ . Such a penalty would require a stochastic model specifying how extra candidate opportunities create spurious scores. Without that model it could erase genuine breadth. Instead, opportunity is handled through a complete declared roster, explicit alpha and q choices, and sensitivity reporting.

9.2 Set rather than multiset

Exact duplicate control occurs before the score vector is formed. A repeated occurrence of the same endpoint–minimal-epitope–normalized-HLA tuple enters once, including recurrence caused by HLA homozygosity or duplicated source rows. The same minimal epitope paired with two distinct HLA alleles forms two tuples. Overlapping but nonidentical epitopes also remain distinct.

This convention prevents literal duplication from manufacturing breadth. It does not assert independence among the distinct tuples. Correlated candidates can still move the aggregate together, and their biological dependence is an interpretive qualification rather than something the Hill number estimates.

10 Interpretive qualifications

Anchor order. Alpha controls the generalized power-mean baseline. It is not a posterior probability that the best candidate is causal. A component–metric-specific alpha remains fixed across policy cohorts after selection.

Hill order. The order q translates a normalized positive-signal distribution into effective breadth. It is not a probability of candidate independence and does not estimate the number of truly causal epitopes.

Gate parameters. The center c is the aggregate $\log-F$ score at which the admitted fraction is one-half; it is not a response-probability threshold. Its meaning depends on the frozen definition of F , logarithmic transformation, and epsilon. The width κ is a score-scale transition width, not a biological variance.

Numerical floor. $\log(10^{-12})$ represents a completed Level 1 zero signal or the declared empty-roster output. It never represents an uncomputed observation in the replacement workflow.

Numerical solution. Equation (20) is solved as a scalar root on the certified interval in Equation (22). Bisection tolerance, maximum iterations, and failure behavior are part of the scoring contract. Naive fixed-point iteration is not part of the definition.

Context and causation. The aggregate does not use cohort identity or peptide length as inputs. That label blindness does not prove that its rankings are invariant to assay geometry, HLA distribution, disease context, or label construction. Transfer must be tested rather than inferred from the form of the equation.

Scope of adaptation. The self-gate limits an offer according to the score it produces. It does not establish that the roster is biologically exhaustive, that distinct tuples are independent, or that a selected parameter tuple will transfer to an unrepresented context.

11 What is and is not frozen

This revision freezes:

- complete- F rather than omission-floor inputs;
- exact tuple-set deduplication;
- the power-mean anchor in Equation (9);
- the floor-aware Hill offer in Equations (17)–(19);
- the self-gated equation and certified solver;
- joint (α, q, c, κ) evaluation;
- worst-cohort-first cross-context selection.

It deliberately does not freeze:

- the production α values;
- any conclusion that COVID genuinely prefers mean after complete- F rescoring;
- the production transport-gate threshold;
- a selected tuple for any component or metric;
- the claim that Level 2 adaptation alone will produce a candidate-conservative survivor.

12 Implementation correspondence

The authoritative implementation is colocated with this memo in `supported_ap_code/crates/select_adaptive_hillq`. It has one complete- F code path and no compatibility mode for the former max-anchored or omission-floor selector. Its manifests record the explicit alpha and Hill grids, the complete formula, the worst-cohort objective, input hashes, solver settings, and selected parameters.